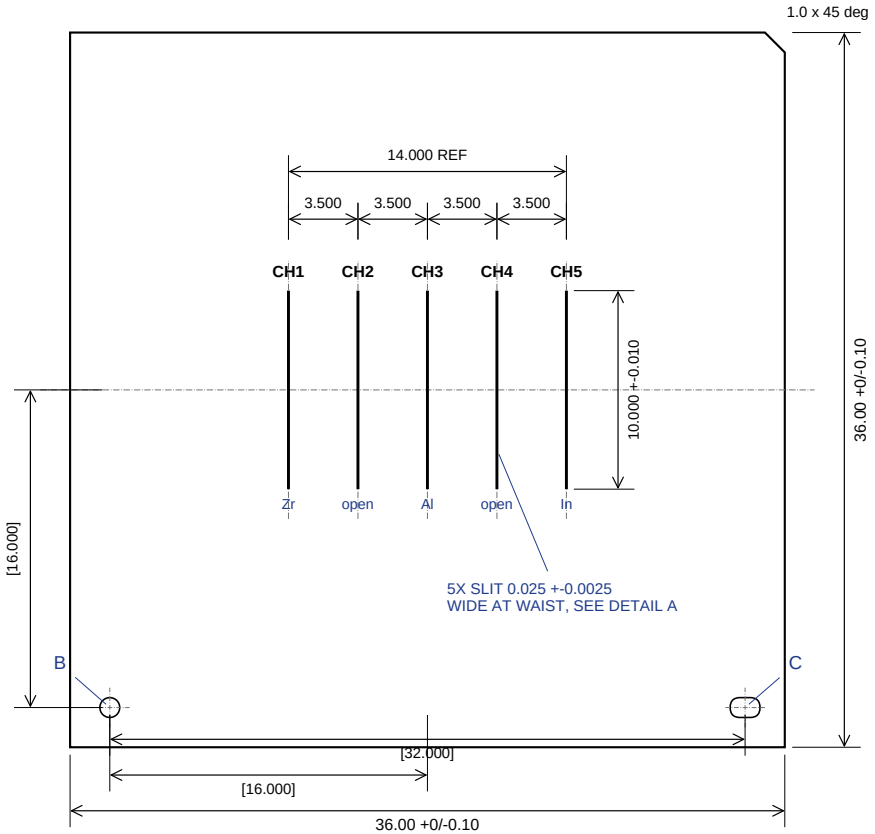
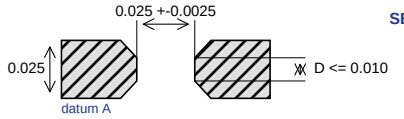
## SP1 PLAN, VIEW FROM SUN SIDE (FACE TOWARD SP2)




[ ] = basic: the pattern center located from datum B, the X axis from B to C.




DETAIL A SECTION ACROSS ONE SLIT
Not to scale. Sun side up, datum A down. The faces open wider than the waist (normal double-sided etch).

## FEATURE TABLE

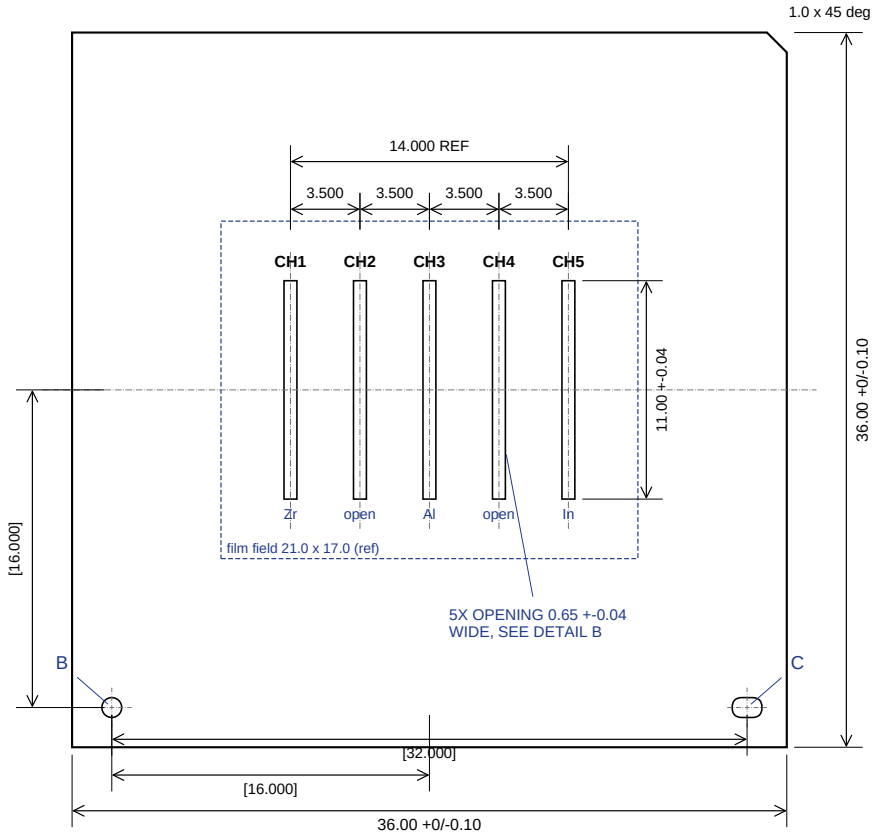
| | |
|---|---|
| **STOCK** | 304 stainless foil 0.025 (0.001 in) +-0.003. Thickness chosen so the 0.025 slit is at the published minimum slot width for photochemical machining (100-125 % of thickness). |
| **OUTLINE** | 36.00 square +0/-0.10; chamfer 1.0 x 45 deg at +X+Y. Outline centered on the pattern within +-0.05. |
| **SLITS** | 5X, through. Width 0.025 +-0.0025 at the narrowest point of the section, over the full length. Length 10.000 +-0.010. Ends square or R0.025 max. |
| **POSITION** | Centerlines at X = -7.000, -3.500, 0.000, +3.500, +7.000 from the pattern center; Y centered. Each centerline within +-0.005 of true position over its whole length (non-cumulative). This also holds the slits parallel. |
| **SECTION** | Symmetric double-sided etch. The opening is narrowest near mid-thickness and stays within the width tolerance over a depth D <= 0.010 (Detail A). D is a maximum: a sharper waist is better. |
| **DATUM A** | The face away from the sun (lower face). It seats on the housing bulkhead. |
| **DATUM B** | Round hole dia 1.000 +0.005/-0 at X -16.000, Y -16.000. |
| **DATUM C** | Slot 1.000 +0.005/-0 wide x 1.500 long, long axis along X, at X 16.000, Y -16.000. B and C within +-0.005 of the slit pattern. Both are for the bonding jig. |
| **FLATNESS** | Free-state flatness not controlled. No kinks, creases or handling dents. |
| **WITNESS** | Etch a witness coupon with three slits of this drawing on the same panel as each lot. Section one coupon across a slit and report the measured profile and D. |
| **REPORT** | Per part, by serial: slit width at 1, 3, 5, 7 and 9 mm along each slit (25 values); the five centerline positions; datum hole sizes. Per lot: the witness section. |

## NOTES

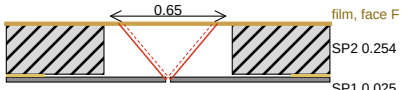
1. Units: millimeters. Dimensions apply to the finished part at 20 C. Tolerances unless stated: +-0.05.
2. Plan views look down from the sun side. The 1.0 x 45 deg chamfer at the upper right sets orientation on every part.
3. SP1, SP2, SP3: AISI 304 stainless, by photochemical machining (double-sided etch). The DXF of the same part number is the geometry for the photo-tool; this drawing sets the tolerances. Material certificate requested.
4. No burrs, slag, oxide or etch residue at any aperture edge. Clean and oil-free. No coating, plating or marking on the part; identify each part by serial number on its package.
5. SP1 and SP2 go to Luxel for film and adhesive bonding and vacuum bake: no oils or silicone-bearing materials in handling or packaging. Pack flat, separated by lint-free tissue.
6. Design authority: Carthage College (K. Crosby, kcrosby@carthage.edu). Questions before cutting metal are welcome.
7. SP1 is the optical aperture of the instrument. Treat the slits as optical surfaces: no contact with tools.

| | |
|---|---|
| PART | **SLANT-SP1 Slit sheet** |
| MATERIAL | AISI 304 stainless foil, 0.025 |
| QTY | per PO |
| MASS (model) | 0.26 g |
| PROJECT | SLANT slit-plate assembly, rev A |
| OWNER | Carthage College, Space Sciences |
| STATUS | Issued for quote |
| DRAWN / CHECKED | from parameters 2026-10-14 / _____ |
| SHEET REV | A |

## SP2 PLAN, VIEW OF FACE F (SUN SIDE, FILMS)




[ ] = basic: the pattern center located from datum B, the X axis from B to C.




DETAIL B SECTION ACROSS ONE CHANNEL
True scale. Red lines: light reaching the slit edges at +-40 deg, the limit of the partition comb. Minimum opening 0.610; 0.555 needed with alignment and slit tolerance. Yellow, under SP2: SP1-SP2 adhesive, <= 0.015 thick, kept 0.30 back from the opening edge.

## FEATURE TABLE

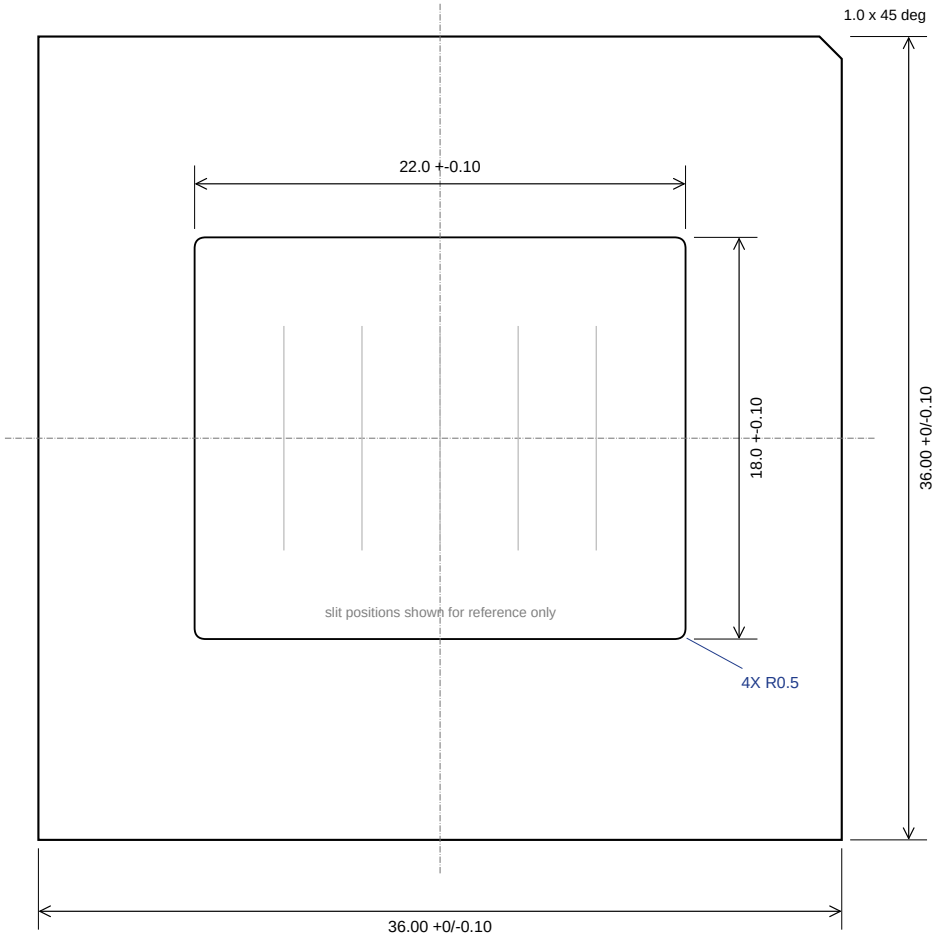
| | |
|---|---|
| **STOCK** | 304 stainless 0.254 (0.010 in) +-0.013. |
| **OUTLINE** | 36.00 square +0/-0.10; chamfer 1.0 x 45 deg at +X+Y, matching SP1. |
| **OPENINGS** | 5X, through. Width 0.65 +-0.04 at the narrowest point of the section. Length 11.00 +-0.04. End corners R0.25 max. |
| **POSITION** | Centerlines at X = -7.000, -3.500, 0.000, +3.500, +7.000 from the pattern center (the same as SP1); Y centered. Each within +-0.010 of true position. |
| **PURPOSE** | Each opening must pass light reaching its SP1 slit at up to 40 deg from normal across the slit and 45 deg along it. With the slit tolerance and +-0.025 assembly alignment the minimum width needed is 0.555 (Detail B). |
| **FACE F** | Sun side. Luxel bonds the films here. Within the film field: no scratches, pits or etch stains visible at 10x. |
| **FACE S** | Bonds to SP1. |
| **DATUM B** | Round hole dia 1.000 +0.04/-0 at X -16.000, Y -16.000. |
| **DATUM C** | Slot 1.000 +0.04/-0 x 1.500, long axis along X, at X 16.000, Y -16.000. Within +-0.010 of the opening pattern. |
| **FLATNESS** | 0.05 over the whole face, after etching. Flatten or stress-relieve at the vendor's discretion. |
| **REPORT** | Per part, by serial: width of each opening at 2, 5.5 and 9 mm along it; the five centerline positions; flatness. |

## NOTES

1. Units: millimeters. Dimensions apply to the finished part at 20 C. Tolerances unless stated: +-0.05.
2. Plan views look down from the sun side. The 1.0 x 45 deg chamfer at the upper right sets orientation on every part.
3. SP1, SP2, SP3: AISI 304 stainless, by photochemical machining (double-sided etch). The DXF of the same part number is the geometry for the photo-tool; this drawing sets the tolerances. Material certificate requested.
4. No burrs, slag, oxide or etch residue at any aperture edge. Clean and oil-free. No coating, plating or marking on the part; identify each part by serial number on its package.
5. SP1 and SP2 go to Luxel for film and adhesive bonding and vacuum bake: no oils or silicone-bearing materials in handling or packaging. Pack flat, separated by lint-free tissue.
6. Design authority: Carthage College (K. Crosby, kcrosby@carthage.edu). Questions before cutting metal are welcome.
7. Face F receives 0.2-0.3 um metal films. Protect it in packing.

| | |
|---|---|
| PART | **SLANT-SP2 Carrier plate** |
| MATERIAL | AISI 304 stainless, 0.254 |
| QTY | per PO |
| MASS (model) | 2.56 g |
| PROJECT | SLANT slit-plate assembly, rev A |
| OWNER | Carthage College, Space Sciences |
| STATUS | Issued for quote |
| DRAWN / CHECKED | from parameters 2026-10-14 / _____ |
| SHEET REV | A |

## SP3  PLAN, VIEW FROM SUN SIDE




## FEATURE TABLE

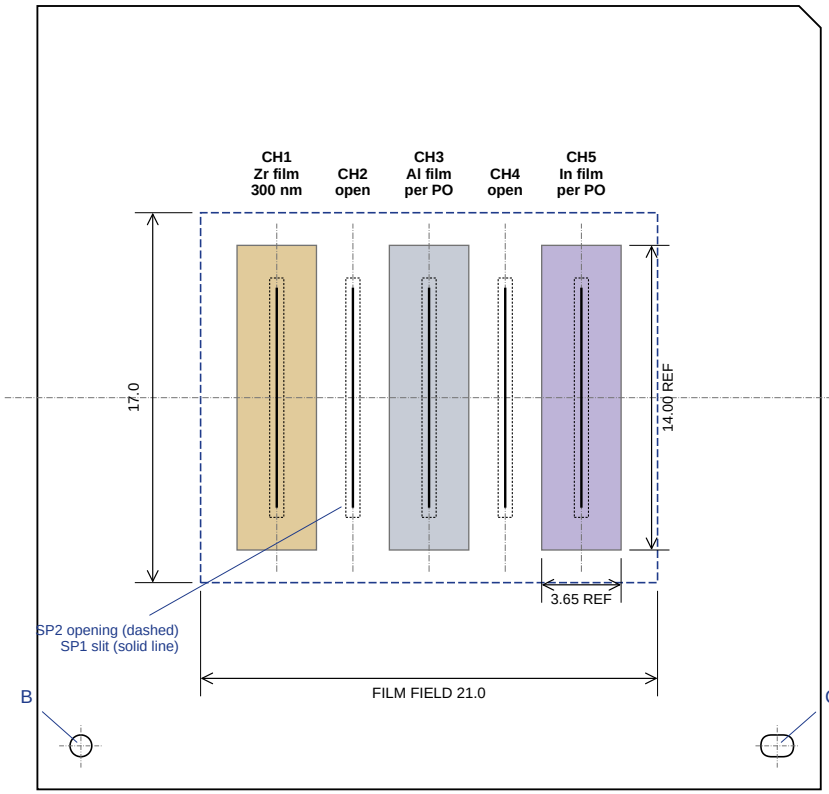
| | |
|---|---|
| **STOCK** | 304 stainless shim, supplied in three thicknesses: 0.178, 0.203, 0.229 (0.007, 0.008, 0.009 in), each +-0.010. Order equal numbers of each. |
| **OUTLINE** | 36.00 square +0/-0.10; chamfer 1.0 x 45 deg at +X+Y. |
| **WINDOW** | 22.0 x 18.0 +-0.10, centered +-0.10, corners R0.5 +-0.2. Larger than the 18.0 x 15.0 window of the H09 cartridge above it, so it never limits the light. |
| **FUNCTION** | Fills the 0.50 deep H09 pocket above the bonded SP1 + SP2 stack, so the housing is unchanged. It lies on SP2 face F outside the film field and is clamped by the H09 flange. Not bonded. |
| **SELECTION** | At integration Carthage measures the pocket depth and the stack height of the bonded assembly, and fits the SP3 that leaves the total 0.000 to +0.025 above the pocket depth. |
| **SUPPLY** | Carthage part; not supplied to Luxel. May be laser cut or etched. |
| **FLATNESS** | 0.05. |

## NOTES

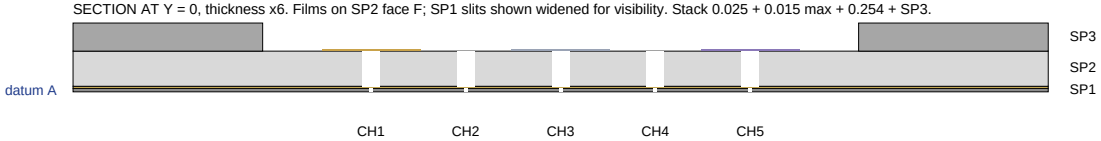
1. Units: millimeters. Dimensions apply to the finished part at 20 C. Tolerances unless stated: +-0.05.
2. Plan views look down from the sun side. The 1.0 x 45 deg chamfer at the upper right sets orientation on every part.
3. SP1, SP2, SP3: AISI 304 stainless, by photochemical machining (double-sided etch). The DXF of the same part number is the geometry for the photo-tool; this drawing sets the tolerances. Material certificate requested.
4. No burrs, slag, oxide or etch residue at any aperture edge. Clean and oil-free. No coating, plating or marking on the part; identify each part by serial number on its package.
5. SP1 and SP2 go to Luxel for film and adhesive bonding and vacuum bake: no oils or silicone-bearing materials in handling or packaging. Pack flat, separated by lint-free tissue.
6. Design authority: Carthage College (K. Crosby, kcrosby@carthage.edu). Questions before cutting metal are welcome.

| | |
|---|---|
| PART | **SLANT-SP3  Spacer shim** |
| MATERIAL | AISI 304 stainless shim, three thicknesses |
| QTY | per PO |
| MASS (model) | 1.46 g (0.203) |
| PROJECT | SLANT slit-plate assembly, rev A |
| OWNER | Carthage College, Space Sciences |
| STATUS | Issued for quote |
| DRAWN / CHECKED | from parameters 2026-10-14  /  _____ |
| SHEET REV | A |

## SPA PLAN, VIEW FROM SUN SIDE (SP3 NOT SHOWN)




SECTION AT Y = 0, thickness x6. Films on SP2 face F; SP1 slits shown widened for visibility. Stack 0.025 + 0.015 max + 0.254 + SP3.




## FEATURE TABLE

**BUILD** Luxel: SP1 bonded to SP2 face S, three films bonded to SP2 face F. Carthage supplies SP1 and SP2 and adds SP3 at integration.

**ALIGN** Every SP1 slit centerline within +-0.025 of its SP2 opening centerline in X and +-0.10 in Y, all five at once. Outlines coincide within +-0.10. B and C give a coarse jig location; final alignment on the slits.

**ADHESIVE** SP1 to SP2: low-outgassing (ASTM E595 TML <= 1.0 %, CVCM <= 0.10 %), Luxel's choice. Bond line <= 0.015. None within 0.30 of any SP2 opening edge, none in or on the slits, none on datum A. Pattern must leave no sealed air pocket between the plates.

**FILMS** One per opening on face F, unsupported (no mesh): CH1 Zr 300 nm; CH3 Al and CH5 In, thickness per the purchase order. CH2 and CH4 stay open: no film, and no adhesive within 0.5 of their opening edges.

**OVERLAP** Each film overlaps its opening edge by >= 1.5 on all sides (>= 1.0 if agreed with Carthage). Nominal patch 3.65 x 14.00, centered on the opening. Epoxy kept back from the opening per Luxel's process.

**FIELD** All film and film adhesive inside the 21.0 x 17.0 film field centered on the pattern. Outside it face F stays bare, because SP3 is clamped there.

**VENTING** The space under each film vents through its slit. Do not seal the slits.

**INSPECT** Per film: Luxel's visible-light pinhole inspection over the whole unsupported area; report method, detection limit and result. Per assembly: photograph of face F and of each slit at >= 50x after cure, showing no adhesive or particles in the slits.

**DATUM** Optical datum: the slit waist plane, 0.0125 above datum A. The housing sets the waist-to-detector distance; nothing on this assembly sets it.

**PACK** Luxel standard filter container, film side protected. Assemblies with indium: dry nitrogen or vacuum storage per Luxel's recommendation.

## NOTES

1. Units: millimeters. Dimensions apply to the finished part at 20 C. Tolerances unless stated: +-0.05.
2. Plan views look down from the sun side. The 1.0 x 45 deg chamfer at the upper right sets orientation on every part.
3. SP1, SP2, SP3: AISI 304 stainless, by photochemical machining (double-sided etch). The DXF of the same part number is the geometry for the photo-tool; this drawing sets the tolerances. Material certificate requested.
4. No burrs, slag, oxide or etch residue at any aperture edge. Clean and oil-free. No coating, plating or marking on the part; identify each part by serial number on its package.
5. SP1 and SP2 go to Luxel for film and adhesive bonding and vacuum bake: no oils or silicone-bearing materials in handling or packaging. Pack flat, separated by lint-free tissue.
6. Design authority: Carthage College (K. Crosby, kcrosby@carthage.edu). Questions before cutting metal are welcome.
7. Parts list: SP1 slit sheet x1, SP2 carrier x1, films x3 (Luxel), adhesive (Luxel); SP3 shim x1 (Carthage, at integration).
8. Channel order is fixed by this drawing: films at CH1, CH3, CH5, 7.0 apart; CH2 and CH4 open.

| | |
|---|---|
| PART | **SLANT-SPA Slit filter assembly** |
| MATERIAL | SP1 + SP2 + Zr, Al, In films (+ SP3 by Carthage) |
| QTY | per PO |
| MASS (model) | about 2.8 g without SP3 |
| PROJECT | SLANT slit-plate assembly, rev A |
| OWNER | Carthage College, Space Sciences |
| STATUS | Issued for quote |
| DRAWN / CHECKED | from parameters 2026-10-14 / _____ |
| SHEET REV | A |